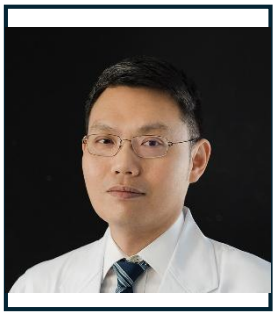
Speaker Information



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Talk Information

Strain-specific Microbe Modulates Stress Response via a Phenylalanine Co-metabolite

Commensal microbiota critically modulate host stress responses, yet specific mechanisms remain elusive. Here, we report that colonizing microbiota-deficient mice with specific *Enterococcus faecalis* strains significantly alleviates stress-induced gut dysmotility. This protective effect operates via the central nervous system, independent of peripheral adrenal or sympathetic pathways, as chemogenetic inhibition of central stress circuits recapitulates the dysmotility phenotype. Utilizing multi-omics, we identified phenylacetylglycine, a microbial-host co-metabolite, as the key mediator. Delivering this metabolite mitigated stress-induced dysmotility, and genomic analyses linked this strain-specific protection to a distinct bacterial enzyme in phenylacetate metabolism. These findings highlight a novel microbiome-metabolite-brain axis by which a single microbe mitigates stress, revealing therapeutic potential for stress-related disorders.

Research Overview (for collaboration discussion)

[Research Interests] Gut-Brain Axis; Host-Microbe Interaction

[Current Research Projects] Decoding Microbial Signals in the Gut-Brain Axis

[Techniques] Microbial manipulations; Chemogenetics; Behaviors

[Aspects or techniques of interest for future collaboration] Open to discuss

Short Bio

Wei-Li Wu, Ph.D. is an Associate Professor in the Department of Physiology, College of Medicine, National Cheng Kung University. He received his Ph.D. in Neuroscience from Academia Sinica. From 2012–2018, he trained as a postdoctoral fellow at Caltech in late Prof. Paul Patterson's and Sarkis Mazmanian's group to study the gut–brain axis and microbial influences on behavior. Since 2018, Dr. Wu has led his own lab at NCKU, focusing on how gut microbiota shape brain circuits and innate behaviors, aiming to uncover microbial signals that influence brain health and inspire microbiota-based therapies.